

DEEPANSHU SHARMA

Apache Flink Developer | Confluent Kafka Engineer | Java Backend Developer

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PROFESSIONAL SUMMARY

Apache Flink and Confluent Kafka focused developer with 4.5 years of experience building Java backend services, CDC pipelines, Kafka/Flink processing flows, and real-time data platforms across banking, insurance, and IoT analytics. Hands-on with Confluent Kafka, Kafka Connect, Debezium and Oracle CDC patterns, Flink SQL, stateful processing, Spring Boot, Redis, SQL, OpenSearch, AWS/GCP, Docker/Kubernetes, and Linux production troubleshooting. Recent work includes Allianz R2 search modernization, banking CDC/Flink PoCs for RAKBANK, ACLEDA Bank and Bank of Commerce contexts, and production platforms for CIMB Bank and Kloudspot. Comfortable moving between code, data modeling, Kafka internals, connector debugging, production incidents, and stakeholder-facing technical documentation.

CORE TECHNICAL SKILLS

Flink	Apache Flink, Flink SQL, event-time windows, stateful processing, stateful joins, enrichment, CEP, checkpointing concepts, backpressure analysis
Kafka / CDC	Confluent Kafka, Apache Kafka, Kafka Connect, Debezium CDC patterns, Confluent Oracle CDC, Oracle XStream/GoldenGate concepts, Schema Registry, Avro/Protobuf awareness, partitions, offsets, retention, compaction
Backend / Data	Java, Spring Boot, REST APIs, multithreading, SQL, PostgreSQL, MySQL, Oracle, Redis, MongoDB, OpenSearch
Cloud / DevOps	AWS EC2/EKS/RDS/S3, AWS Managed Service for Apache Flink, GCP, Docker, Kubernetes, Jenkins CI/CD, Linux

PROFESSIONAL EXPERIENCE

Kafka & Flink Streaming Consultant | SoftwareOne / Allianz Technology | Apr 2026 - Present | Freelance / Insurance Technology

- Contributing to Allianz R2 Summary Search modernization, shifting high-latency query-time joins into an ingestion-time materialized view pipeline.
- Designed and validated CDC-to-search architecture: Aurora PostgreSQL/RDS CDC -> Kafka Connect on AWS EKS -> Allianz EBS/Confluent Kafka -> AWS Managed Apache Flink -> Amazon OpenSearch.
- Modeled CONTRACT_ID as the business key for Kafka partitioning, Flink keyed state, entity reconstruction, and OpenSearch document identity.
- Defined component boundaries: Kafka Connect/Debezium for CDC and light transformations; Flink SQL for stateful joins, enrichment, recomputation, late/out-of-order handling, and snapshot emission.
- Prepared HLD/deck material for stakeholders covering managed Flink choices, source DB load risk, replay/rebuild strategy, OpenSearch document design, and operational ownership.

Senior Kafka & Flink Developer | Optimum InfoSystem - Client: RAKBANK UAE / Banking PoCs | Mar 2026 - Present

- Building banking-grade Kafka/Flink solutions and PoCs around Oracle/PostgreSQL CDC, Confluent Kafka, Kafka Connect, and Flink SQL.
- Worked on RAKBANK-related streaming and CDC implementation planning, connector setup, topic/config validation, networking/DNS troubleshooting, and production-readiness checks.
- Supported ACLEDA Bank PoC using Oracle XE 21c, local Kafka Connect, Confluent Cloud Kafka, and Flink SQL; analyzed Debezium-style CDC payloads for T24 XML event flows.
- Handled ACLEDA CLOB/LOB CDC considerations by mapping XML payload handling through Confluent Oracle CDC LOB topics and downstream Flink SQL parsing/enrichment design.
- Worked on Bank of Commerce / PRIME banking CDC flow: Oracle DB -> Oracle XStream/GoldenGate CDC concepts -> Kafka/Confluent Cloud -> Flink transformation -> filtered Kafka topic -> JDBC sink/PostgreSQL SMS Interceptor.
- Converted banking requirements into pipeline steps, demos, failure checks, and success criteria that non-streaming stakeholders could understand.

Senior Data Streaming Engineer / Kafka Specialist | Customer Centria - Client: CIMB Bank Malaysia | May 2024 - Aug 2025

- Built and supported high-throughput data pipelines using Confluent Kafka, Apache Flink, Redis, SQL, and Spring Boot for the CIMB OCTO banking ecosystem.
- Worked across topic design, partition strategy, consumer group behavior, producer/consumer tuning, ACLs, retention planning, lag analysis, and production troubleshooting.
- Implemented Flink event-processing patterns including event-time windows, stateful enrichment, and behavior/event pattern detection for engagement and analytics use cases.
- Developed Spring Boot APIs and SQL-backed services supporting broadcast and targeted mobile delivery, dashboards, campaign analytics, CRM flows, and real-time user engagement.
- Supported large-volume banking streams with prior resume metrics of 50M+ events/day and sub-50ms latency goals in mission-critical environments.
- Improved reliability through partition/parallelism tuning, Redis-backed low-latency access, and cross-team incident analysis.

Data Engineer - Real-Time Analytics | Kloudspot Inc. | Feb 2022 - Apr 2024 | IoT Analytics / Hospitality

- Built Kafka and Flink pipelines over IoT sensor streams including WiFi, Bluetooth, and camera-derived events for real-time analytics products.
- Developed Flink jobs for event-time windows, stateful enrichment, aggregations, and near-real-time dashboard metrics.
- Built Java/Spring Boot microservices and REST APIs for hospitality analytics clients in Japan and Dubai.
- Optimized MongoDB aggregations and indexing paths, reducing query latency from approximately 40 seconds to 10 seconds in analytics workflows.
- Managed Kafka and Redis components in Docker/Kubernetes-based environments and supported Linux production troubleshooting.
- Delivered a P2P video conferencing backend module that reduced dependency on third-party licensing for a product capability.

SELECTED BANKING / INSURANCE PROJECT HIGHLIGHTS

- **Allianz R2 Summary Search:** Aurora PostgreSQL CDC -> Kafka/EBS -> Flink SQL stateful entity reconstruction -> OpenSearch serving index, with CONTRACT_ID as the document/business key.
- **ACLEDA Bank PoC:** Oracle CDC and CLOB/XML handling using Kafka Connect, Confluent Cloud Kafka, LOB topic design, and Flink SQL transformation for T24 event-style data.
- **Bank of Commerce / PRIME CDC:** Oracle XStream/GoldenGate CDC concepts, Kafka/Confluent Cloud transport, Flink filtering/transformation, and JDBC sink to PostgreSQL SMS Interceptor.
- **CIMB OCTO Banking Platform:** Kafka, Flink, Redis, Spring Boot, SQL-backed services, CRM/campaign analytics, broadcast/targeted mobile engagement, and production troubleshooting.
- **Kloudspot IoT Analytics:** Kafka/Flink pipelines for sensor events, event-time aggregation, Spring Boot services, MongoDB optimization, Redis, Docker/Kubernetes, and Linux operations.

ENGINEERING PRACTICES

- Builds streaming pipelines around correctness first: stable keys, ordering assumptions, replayability, idempotent writes, and clear recovery paths.
- Treats Kafka Connect, Flink, and sinks as separate responsibilities, keeping CDC ingestion, stateful transformation, and serving/indexing boundaries clean.
- Debugs beyond application code by reading connector logs, Kafka client behavior, consumer lag, broker/network signals, Linux process state, and downstream symptoms.
- Writes documentation and diagrams that make technical trade-offs understandable for architects, managers, and implementation teams.
- Comfortable with production realities: schema changes, late events, LOB handling, backpressure, DNS/network issues, DLQs, retries, and controlled rebuilds.

EDUCATION & CERTIFICATION

- B.Tech in Computer Science & Engineering - Madhav Institute of Technology and Science, 2022
- Confluent Certified Administrator for Apache Kafka